MAT 3007 – Optimization

Newton's Method

Lecture 24

Instructor: Junchi Yang

CUHK-Shenzhen

Recap

Gradient Descent Method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k).$$

Gradient Descent Method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k).$$

- Constant step size: $\alpha_k = \bar{\alpha}$.

Gradient Descent Method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k).$$

- Constant step size: $\alpha_k = \bar{\alpha}$.
- Exact line search: find α_k that

$$\min_{\alpha} f(x^k - \alpha \nabla f(\boldsymbol{x}^k))$$

Gradient Descent Method:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - \alpha_k \nabla f(\boldsymbol{x}^k).$$

- Constant step size: $\alpha_k = \bar{\alpha}$.
- Exact line search: find α_k that

$$\min_{\alpha} f(x^k - \alpha \nabla f(\boldsymbol{x}^k))$$

- Armijo Line Search: Fix $\gamma \in (0, 1)$. Search step size α_k in $\{1, \sigma, \sigma^2, \sigma^3, \dots\}$, where $\sigma \in (0, 1)$, one by one until we find one that satisfies

$$f(\boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k) - f(\boldsymbol{x}^k) \leq -\gamma \alpha_k \cdot \|\nabla f(\boldsymbol{x}^k)\|^2.$$

Lipschitz Continuity and Strong Convexity

Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\ \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

- If $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable, then it is Lipschitz smooth if and only if $\|\nabla^2 f(\boldsymbol{x})\| \leq L$ for any $\boldsymbol{x} \in \mathbb{R}^n$.

Lipschitz Continuity and Strong Convexity

Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

- If $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable, then it is Lipschitz smooth if and only if $\|\nabla^2 f(\boldsymbol{x})\| \leq L$ for any $\boldsymbol{x} \in \mathbb{R}^n$.

Definition: Strongly Convex Function

A function $f : \Omega \to \mathbb{R}$ is strongly convex with parameter $\mu > 0$ if, for all $\boldsymbol{x}, \boldsymbol{y} \in \Omega$, it satisfies the inequality:

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mu}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2,$$

Lipschitz Continuity and Strong Convexity

Definition: Lipschitz Continuity and Smoothness

A function f is Lipschitz smooth over Ω if its gradient ∇f is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|, \quad \forall\, \boldsymbol{x}, \boldsymbol{y} \in \Omega,$$

- If $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable, then it is Lipschitz smooth if and only if $\|\nabla^2 f(\boldsymbol{x})\| \leq L$ for any $\boldsymbol{x} \in \mathbb{R}^n$.

Definition: Strongly Convex Function

A function $f : \Omega \to \mathbb{R}$ is strongly convex with parameter $\mu > 0$ if, for all $\boldsymbol{x}, \boldsymbol{y} \in \Omega$, it satisfies the inequality:

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) + \frac{\mu}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2,$$

- If $f : \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable, then it is strongly convex if and only if the smallest eigenvalues $\lambda_{\min}(\nabla^2 f(\boldsymbol{x})) \geq \mu$ for any $\boldsymbol{x}$.

Remark: If a function f is L-Lipschitz smooth and μ-strongly convex, we should always have $L \geq \mu$.

Theorem: Global Convergence to Stationary Points



Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method for solving

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \mathbb{R}^n$$

with one of the following step size strategies:

- Exact line search,
- Armijo line search (backtracking) with $\sigma, \gamma \in (0, 1)$.

Theorem: Global Convergence to Stationary Points



Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method for solving

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \mathbb{R}^n$$

with one of the following step size strategies:

- Exact line search,
- Armijo line search (backtracking) with $\sigma, \gamma \in (0, 1)$.

Then, $\{f(\boldsymbol{x}^k)\}_k$ is nonincreasing and every accumulation point of $\{\boldsymbol{x}^k\}_k$ is a stationary point of f.

Theorem: Global Convergence to Stationary Points



Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable and let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method for solving

$$\min_{\boldsymbol{x}} \; f(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \mathbb{R}^n$$

with one of the following step size strategies:

- Exact line search,
- Armijo line search (backtracking) with $\sigma, \gamma \in (0, 1)$.

Then, $\{f(\boldsymbol{x}^k)\}_k$ is nonincreasing and every accumulation point of $\{\boldsymbol{x}^k\}_k$ is a stationary point of f.

Remark: See Proposition 1.2.1 in "Bertsekas, D. P., Nonlinear programming, 3rd Edition." for a proof.

Theorem: Rates for Strongly Convex Problems

Let f be strongly convex with constant μ and Lipschitz smooth with constant L. Let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method and let $\boldsymbol{x}^*$ be the global minimizer of $\min_{\boldsymbol{x}} f(\boldsymbol{x})$.

Theorem: Rates for Strongly Convex Problems

Let f be strongly convex with constant μ and Lipschitz smooth with constant L. Let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method and let $\boldsymbol{x}^*$ be the global minimizer of $\min_{\boldsymbol{x}} f(\boldsymbol{x})$.

With constant step size $\bar{\alpha} \in \left(0, \frac{2}{L}\right)$, we have $\{\boldsymbol{x}^k\}_k$ converges linearly to $\boldsymbol{x}^*$. When $\bar{\alpha} = \frac{2}{L+\mu}$,

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L-\mu}{L+\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|.$$

Theorem: Rates for Strongly Convex Problems

Let f be strongly convex with constant μ and Lipschitz smooth with constant L. Let $\{\boldsymbol{x}^k\}_k$ be generated by the gradient descent method and let $\boldsymbol{x}^*$ be the global minimizer of $\min_{\boldsymbol{x}} f(\boldsymbol{x})$.

With constant step size $\bar{\alpha} \in \left(0, \frac{2}{L}\right)$, we have $\{\boldsymbol{x}^k\}_k$ converges linearly to $\boldsymbol{x}^*$. When $\bar{\alpha} = \frac{2}{L+\mu}$,

convergence of iterates

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L-\mu}{L+\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|.$$

Moreover, with constant step size $\bar{\alpha} \in \left(0, \frac{2}{L}\right)$, exact line search or Armijo line search, $\{f(\boldsymbol{x}^k)\}_k$ converge linearly to $f(\boldsymbol{x}^*)$ (we will specify η in the next slide):

convergence of function value.

$$f(\boldsymbol{x}^{k+1}) - f(\boldsymbol{x}^*) \leq \eta \cdot [f(\boldsymbol{x}^k) - f(\boldsymbol{x}^*)].$$

Convergence Rate: Remarks

Remarks:

- The constant η depends on the step size:

$$\eta = \begin{cases} 1 - 2\mu\bar{\alpha}(1 - \frac{L\bar{\alpha}}{2}) & \text{constant step size: } \bar{\alpha} \in (0, \frac{2}{L}), \\ 1 - \frac{\mu}{L} & \text{exact line search,} \\ 1 - 2\mu\gamma \min\{1, \frac{2\sigma(1-\gamma)}{L}\} & \text{Armijo line search.} \end{cases}$$

step size should be small enough.

need to know L.

Convergence Rate: Remarks

Remarks:

- The constant η depends on the step size:

$$\eta = \begin{cases} 1 - 2\mu\bar{\alpha}(1 - \frac{L\bar{\alpha}}{2}) & \text{constant step size: } \bar{\alpha} \in (0, \frac{2}{L}), \\ 1 - \frac{\mu}{L} & \text{exact line search,} \\ 1 - 2\mu\gamma \min\{1, \frac{2\sigma(1-\gamma)}{L}\} & \text{Armijo line search.} \end{cases}$$

- The optimal rate is achieved for the constant step size $\bar{\alpha} = \frac{1}{L}$ or exact line search. In this case, we obtain:

$$\eta = 1 - \frac{\mu}{L} = 1 - \frac{1}{\kappa} \quad \text{where} \quad \boxed{\kappa = \frac{L}{\mu}.}$$

minimize over $\bar{\alpha}$

$\rightarrow \bar{\alpha} = \frac{1}{L}$

condition number ≥ 1

when κ is larger, the convergence is slower.

More Remarks

Remarks:

- Convexity is essential for ensuring convergence to the global solution.

More Remarks

Remarks:

- Convexity is essential for ensuring convergence to the global solution.
- Lipschitz smoothness and strong convexity are not required for general convergence rates but are critical for linear convergence. Different assumptions lead to different convergence rates.

More Remarks

Remarks:

- Convexity is essential for ensuring convergence to the global solution.
- Lipschitz smoothness and strong convexity are not required for general convergence rates but are critical for linear convergence. Different assumptions lead to different convergence rates.
- For a convex function f that is Lipschitz smooth with constant L, gradient descent with step size $\bar{\alpha} = \frac{1}{L}$ converges as:

$$f(\boldsymbol{x}^k) - f(\boldsymbol{x}^*) \leq \frac{2L\|\boldsymbol{x}^0 - \boldsymbol{x}^*\|^2}{k},$$

which is slower than linear convergence.

Linear Convergence

Definition: Linear Convergence

We say that $\{\boldsymbol{x}^k\}_k$ converges linearly with rate $\eta \in (0,1)$ to $\boldsymbol{x}^* \in \mathbb{R}^n$ if there exists $\ell \geq 0$ such that for all $k \geq \ell$

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \eta \cdot \|\boldsymbol{x}^k - \boldsymbol{x}^*\|.$$

Definition: Linear Convergence

We say that $\{\boldsymbol{x}^k\}_k$ converges linearly with rate $\eta \in (0,1)$ to $\boldsymbol{x}^* \in \mathbb{R}^n$ if there exists $\ell \geq 0$ such that for all $k \geq \ell$

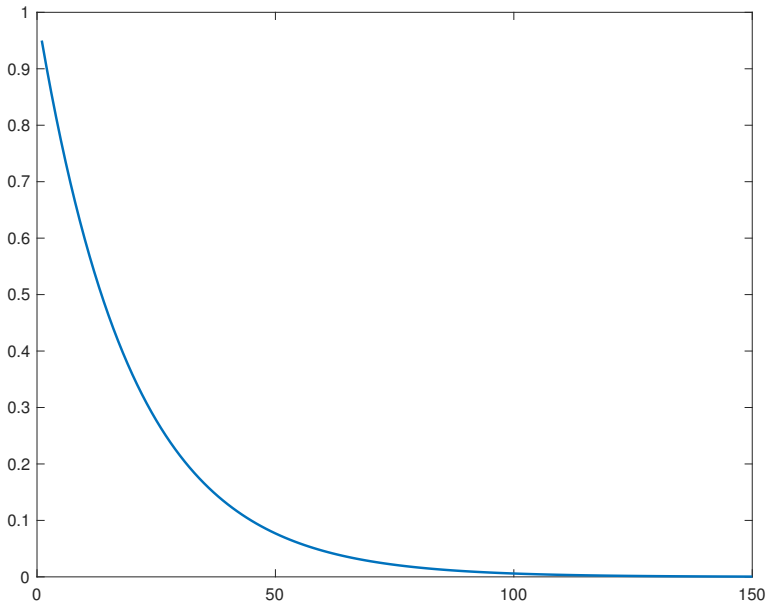
$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \eta \cdot \|\boldsymbol{x}^k - \boldsymbol{x}^*\|.$$

Example:

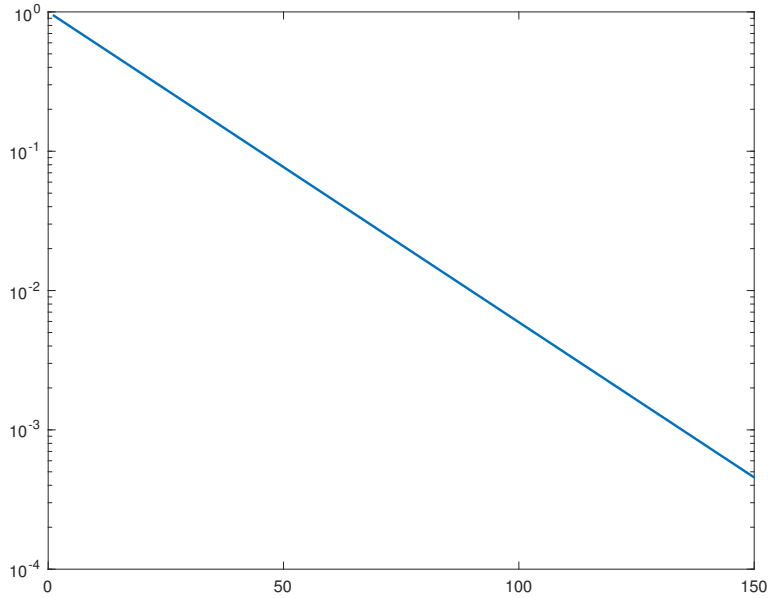
- Let $\rho \in (0,1)$ be given, then the sequence $\{x^k\}_k$ with $x^k := \rho^k$ converges linear to $x^* = 0$ with rate ρ:

$$\frac{|x^{k+1} - x^*|}{|x^k - x^*|} = \frac{\rho^{k+1}}{\rho^k} = \rho, \quad \forall\ k \geq 0.$$

Illustration of Linear Convergence



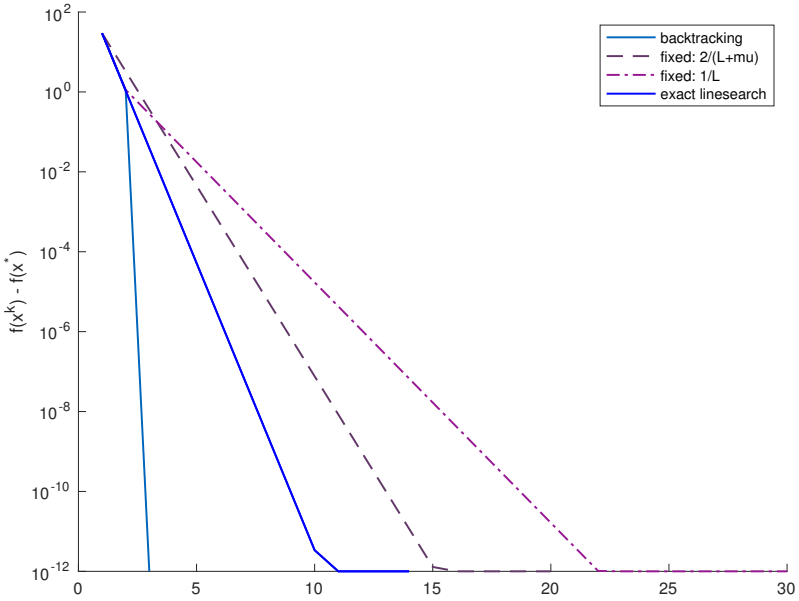
(a) Plot of $\{0.95^k\}_k$

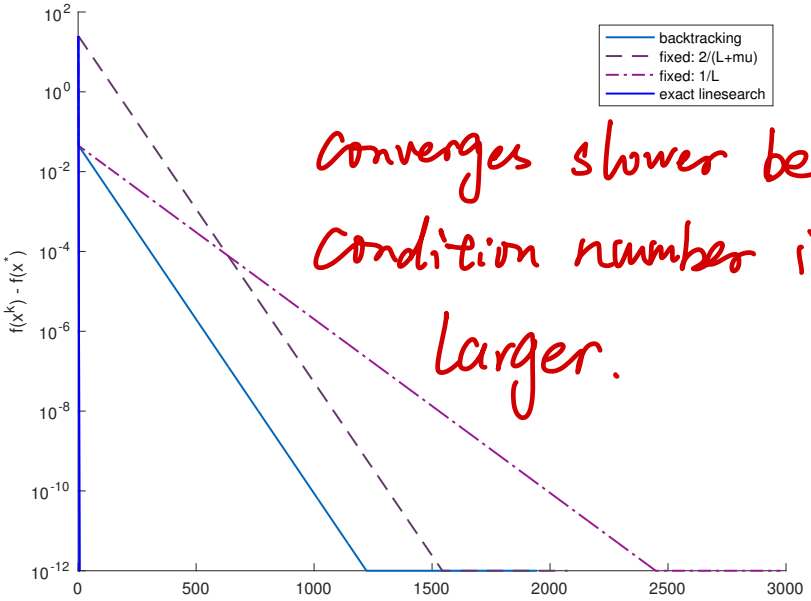


(b) Logarithmic plot of $\{0.95^k\}_k$

- The plot in (b) shows convergence of the adjusted sequence $\tilde{x}^k = \log_{10}(0.95^k) = \log_{10}(0.95) \cdot k \approx -0.022 \cdot k$.
- The labels of the y-axis are given by $10^{\tilde{x}^k}$.
- In logarithmic plots, linear convergence corresponds to (or below) a linear line in k .

Example: Convergence Rates





converges slower because condition number is larger.

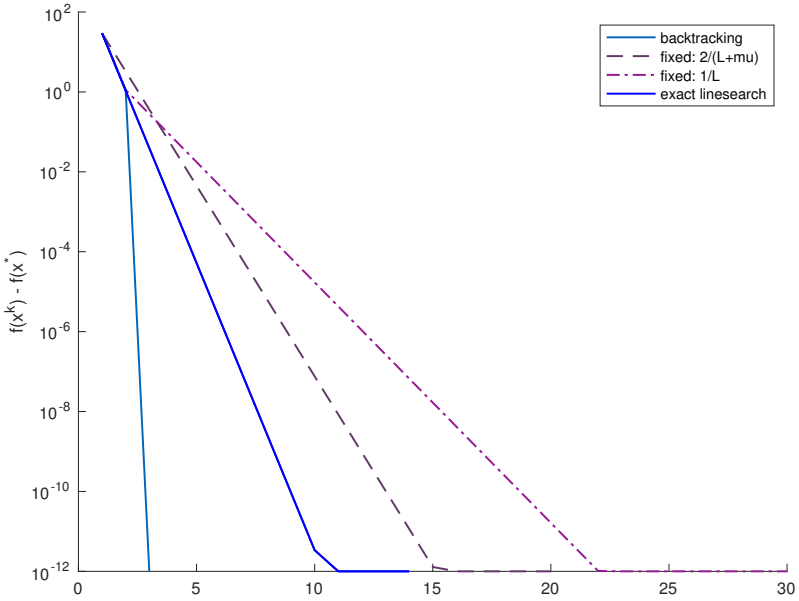
- Comparison of the gradient method with backtracking ($\gamma = \sigma = \frac{1}{2}$) and constant step size for

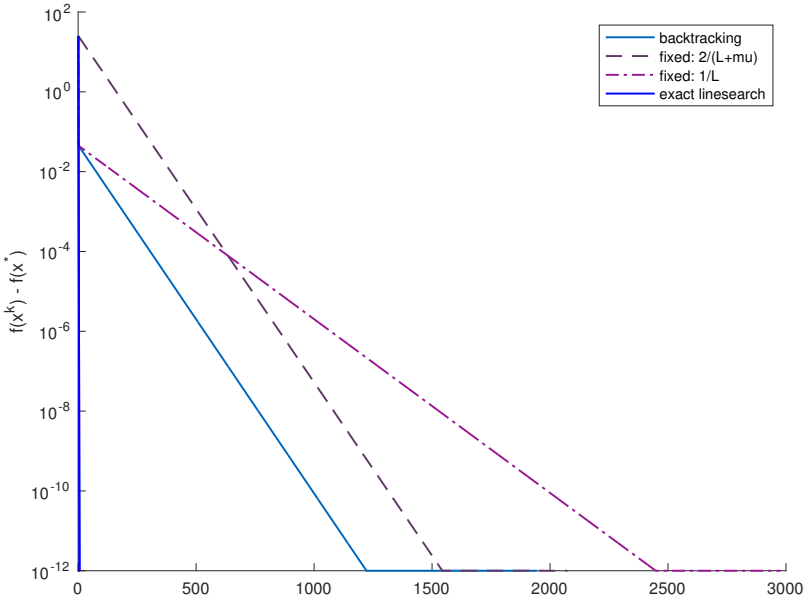
$$\min_{\boldsymbol{x}} \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{A}_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{A}_2 = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{100} \end{pmatrix}.$$

For problem with A_1: $L = 2$, $\mu = 1$ $\Rightarrow$ $\kappa = \frac{L}{\mu} = 2$

A_2 : $L = 2$, $\mu = \frac{1}{100}$ $\Rightarrow$ $\kappa = 200$

Example: Convergence Rates





- Comparison of the gradient method with backtracking ($\gamma = \sigma = \frac{1}{2}$) and constant step size for

$$\min_{\boldsymbol{x}} \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x}, \quad \boldsymbol{A}_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{A}_2 = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{100} \end{pmatrix}.$$

- The condition number is 2 for the problem with $\boldsymbol{A}_1$ and 200 with $\boldsymbol{A}_2$.

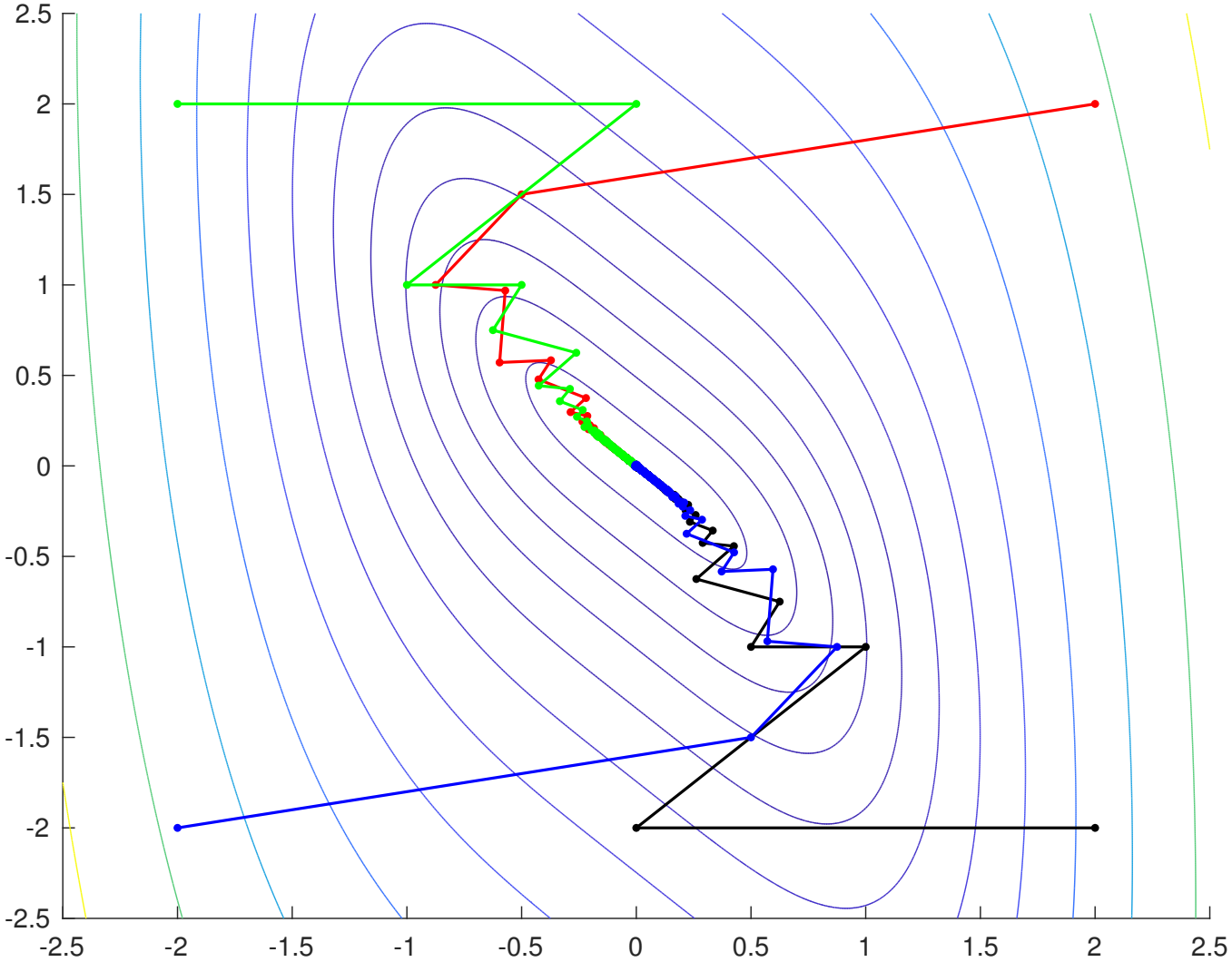
Example: A Convex Problem

We consider the two-dimensional optimization problem

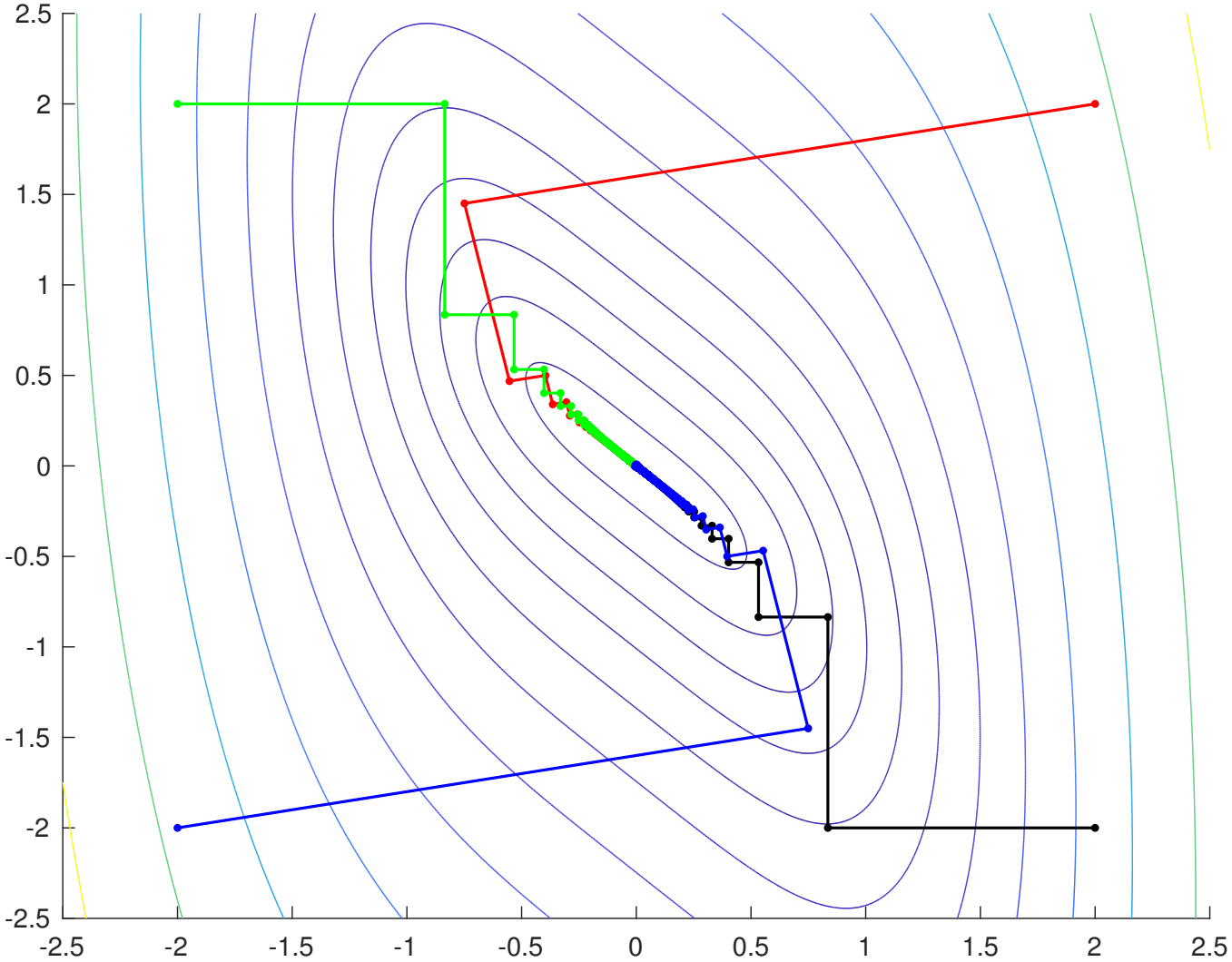
$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \; f(\boldsymbol{x}) = x_1^4 + x_1^2 + x_2^2 + 2x_1x_2 + \frac{1}{4}.$$

- This problem has only one stationary point: $\boldsymbol{x}^* = (0,0)^\top$ (which is a strict global minimizer).
- f is convex but not strongly convex. For this problem, gradient descent does not have a linear convergence.

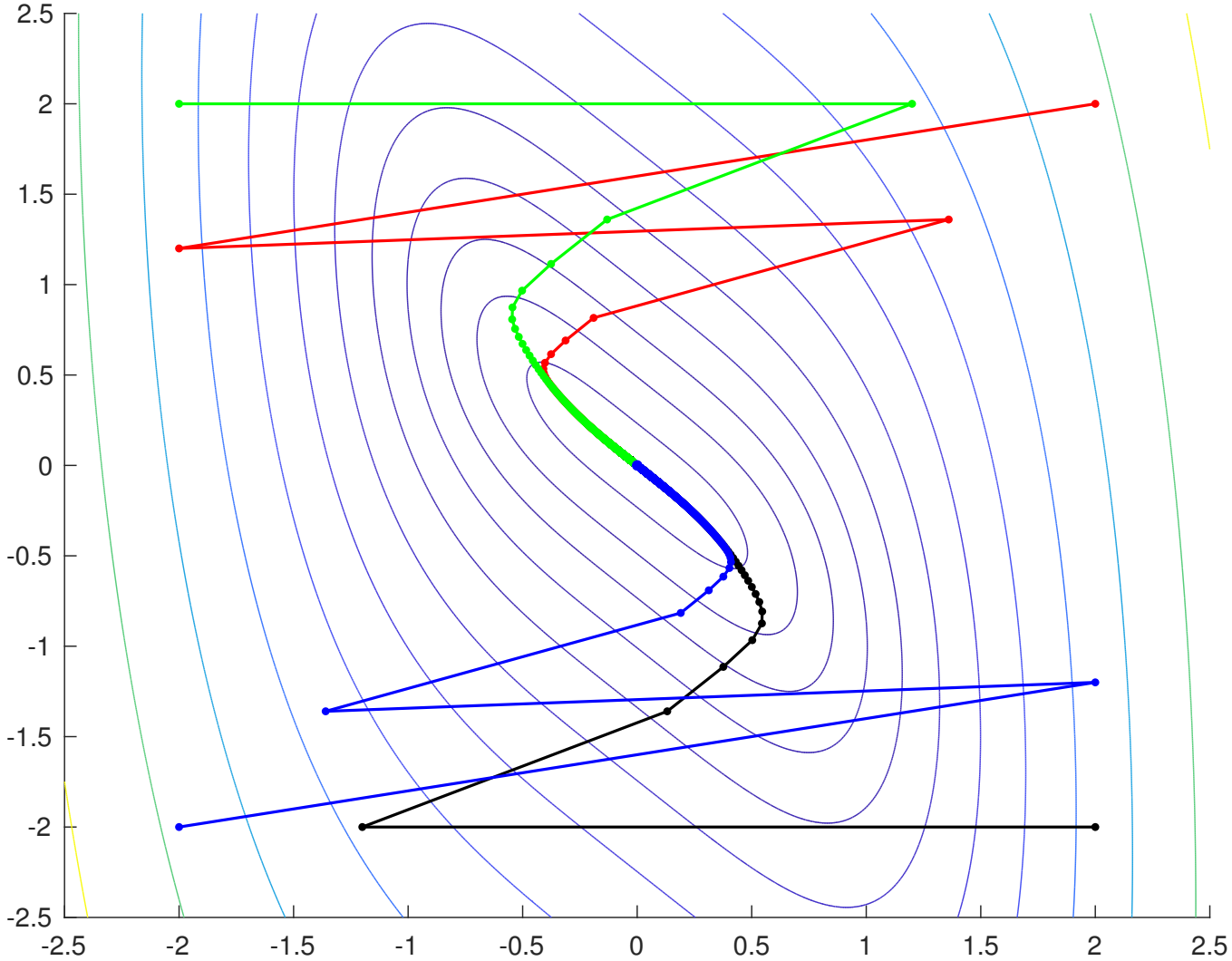
Solution Paths: Backtracking ($\sigma = 0.5$, $\gamma = 0.1$)



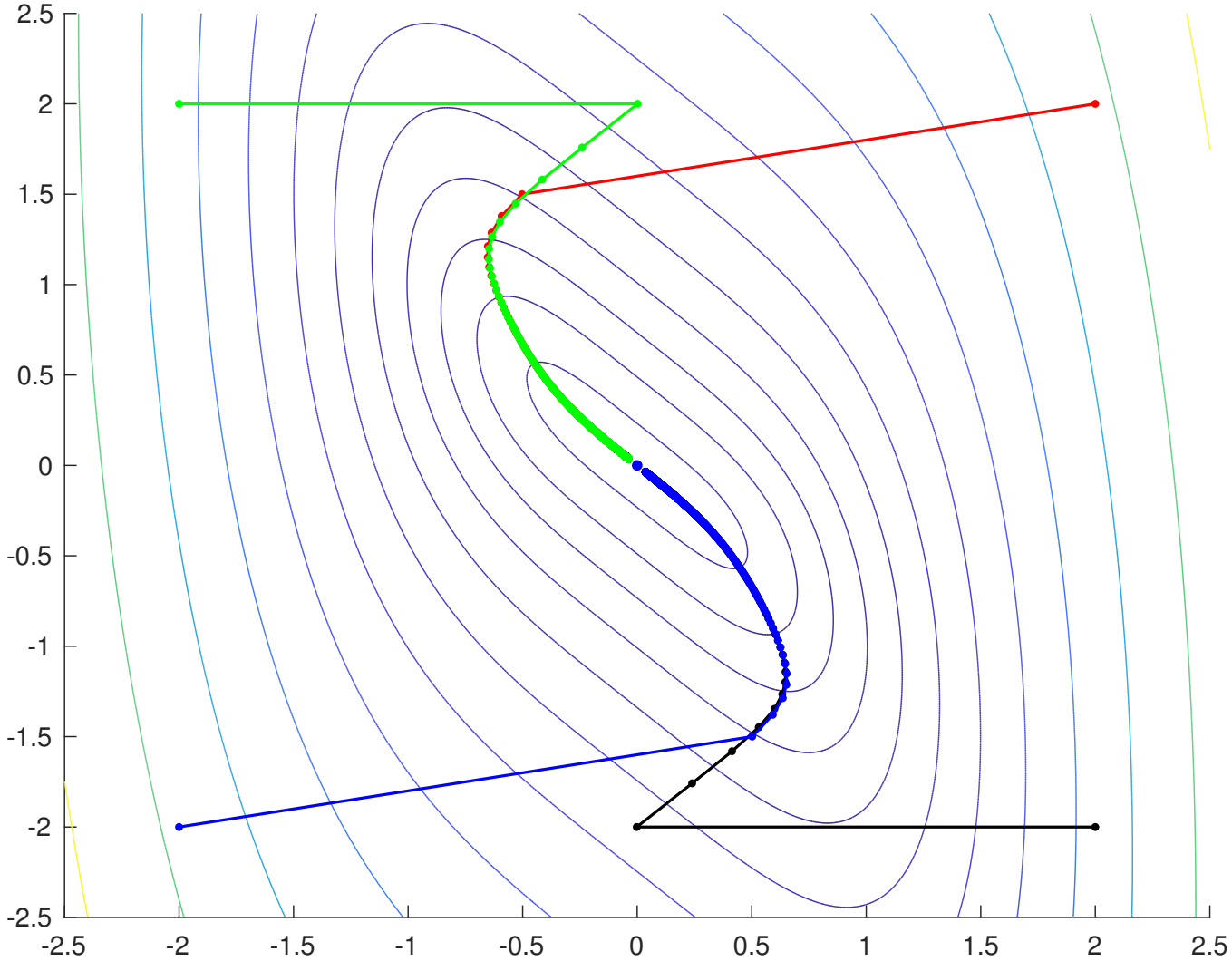
Solution Paths: Exact Line Search



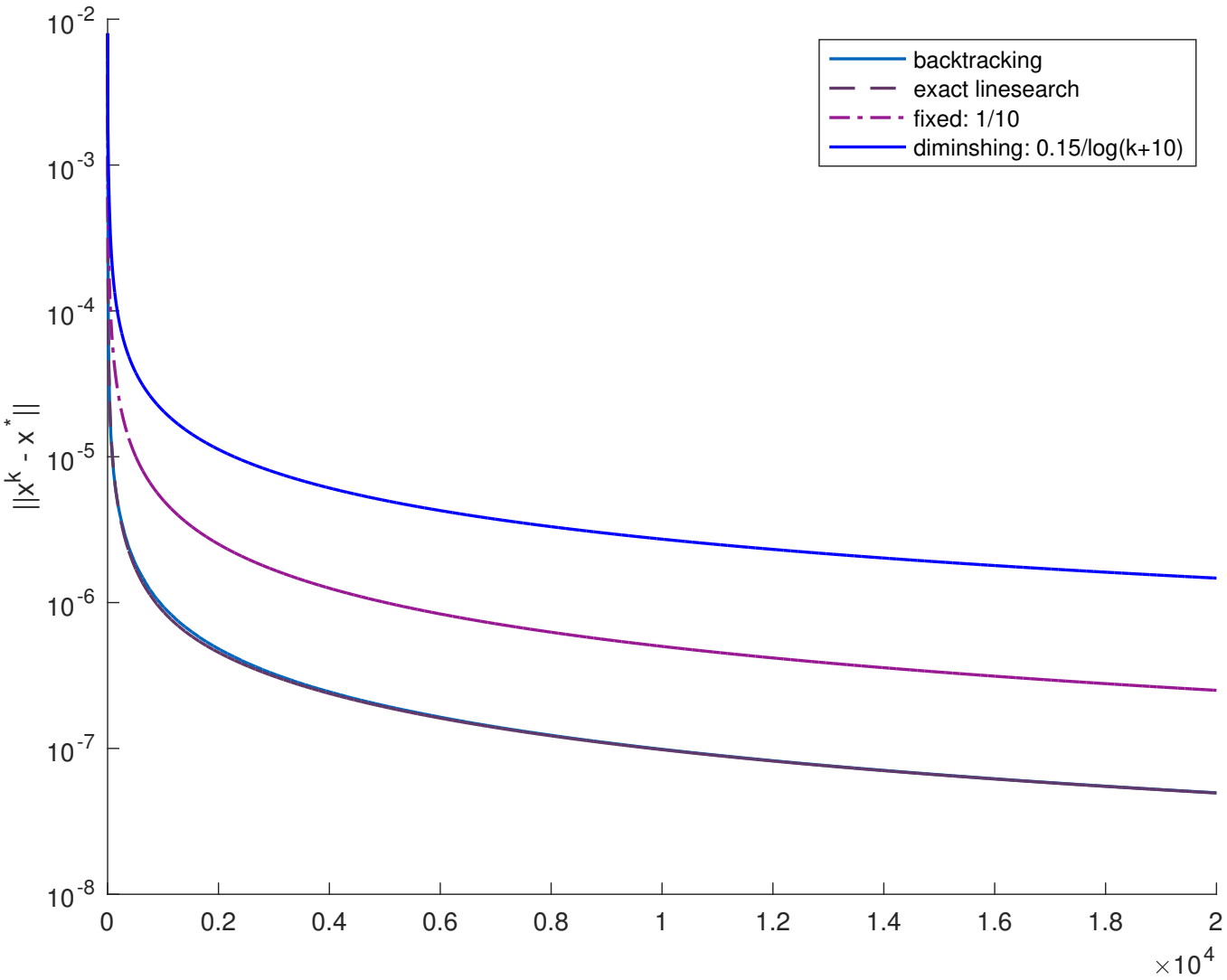
Solution Paths: Constant Step Size $\bar{\alpha} = 1/10$



Solution Paths: Diminishing Steps $0.15/\log(k+10)$



Convergence: Comparison



Discussion and Summary

Pros:

- Only use the first-order information (gradient).
- Global convergence.

Cons:

- Convergence speed may not be fast enough.

Newton's Method — in $\mathbb{R}$

Newton's Method: One Dimension

We want to minimize f:

- A necessary condition is $g(x) = f'(x) = 0$. We first try to find such points.

Newton's Method: One Dimension

We want to minimize f:

- A necessary condition is $g(x) = f'(x) = 0$. We first try to find such points.

First-order Taylor expansion at x^k:

$$g(x) \approx g(x^k) + g'(x^k)(x - x^k)$$

Newton's Method: One Dimension

We want to minimize f:

- A necessary condition is $g(x) = f'(x) = 0$. We first try to find such points.

First-order Taylor expansion at x^k:

$$g(x) \approx g(x^k) + g'(x^k)(x - x^k)$$

Set the right-hand side to be 0 and solve it:

$$x = x^k - \frac{g(x^k)}{g'(x^k)}$$

We choose this x as our next iterate x^{k+1}. Here we assume $g'(x) \neq 0$ at each step.

Illustration of Newton's Method to Find $g(x) = 0$

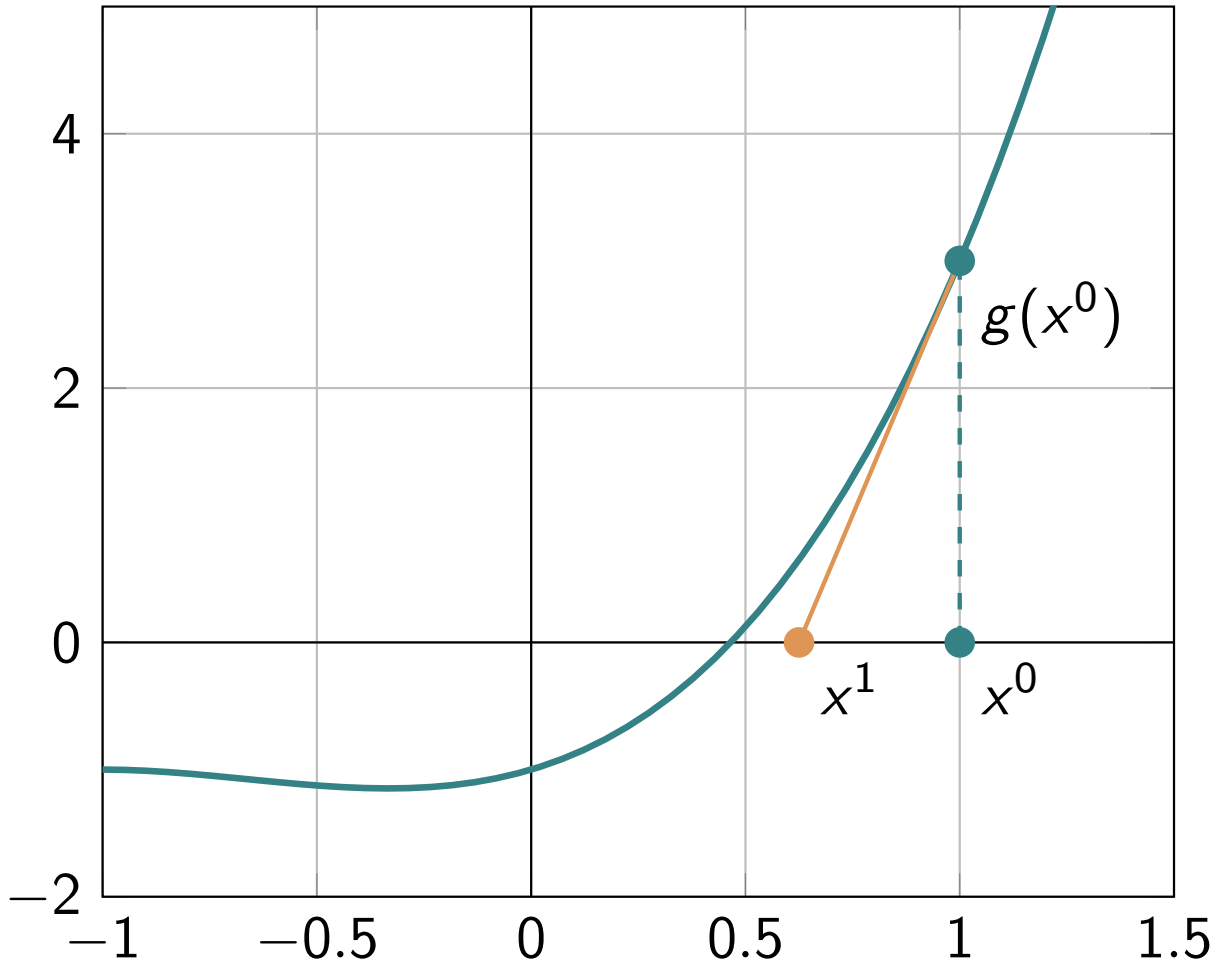


Illustration of Newton's Method to Find $g(x) = 0$

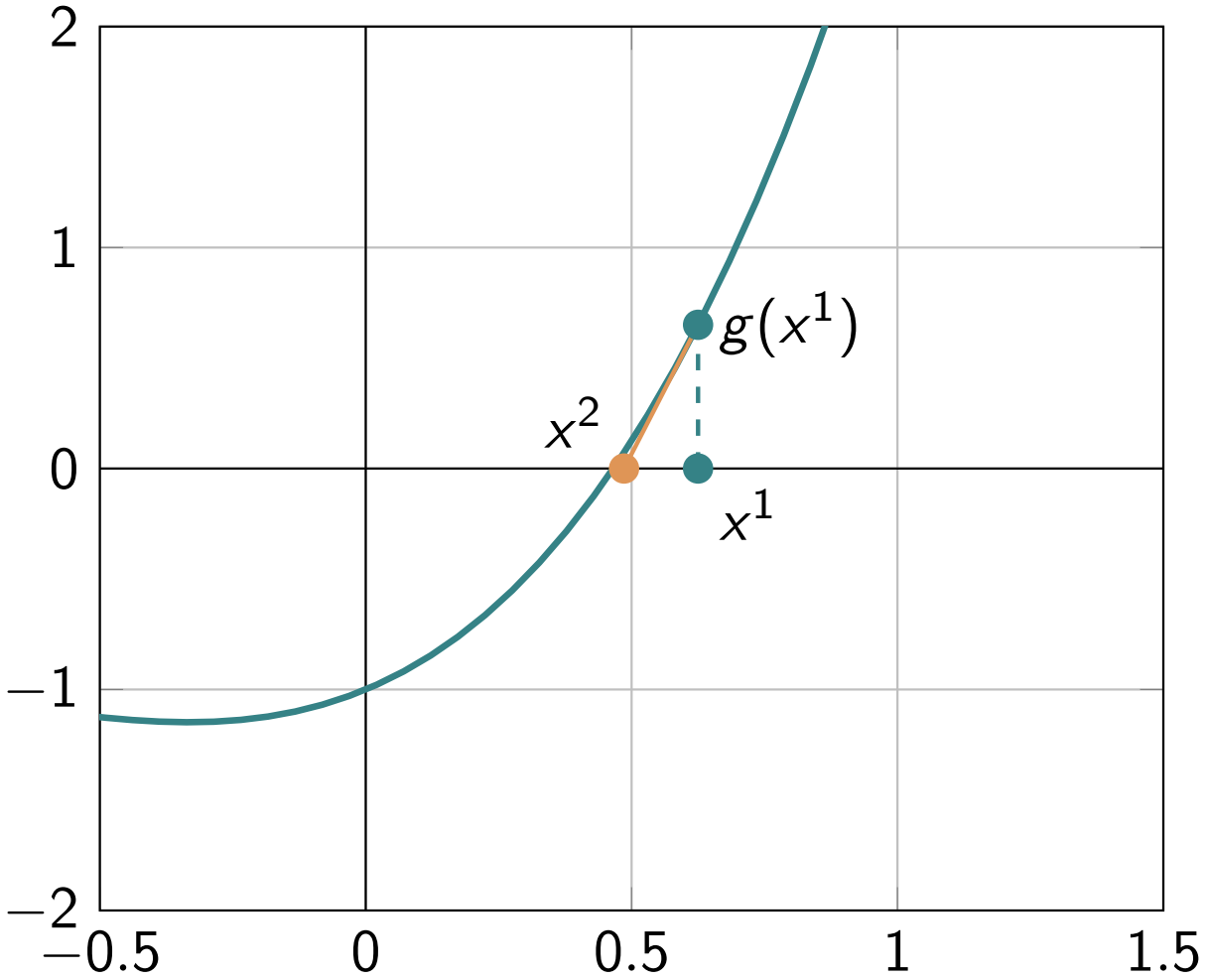
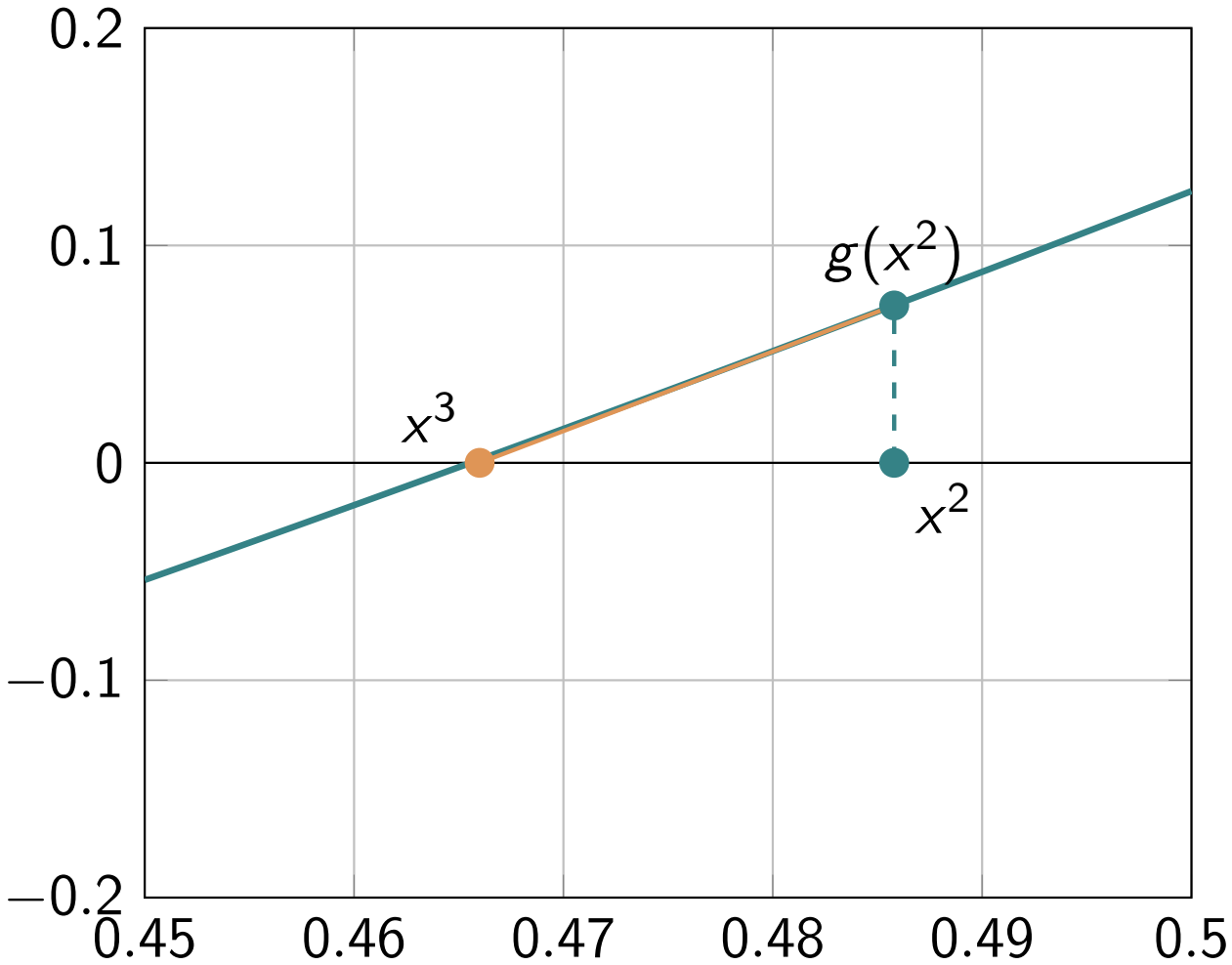


Illustration of Newton's Method to Find $g(x) = 0$

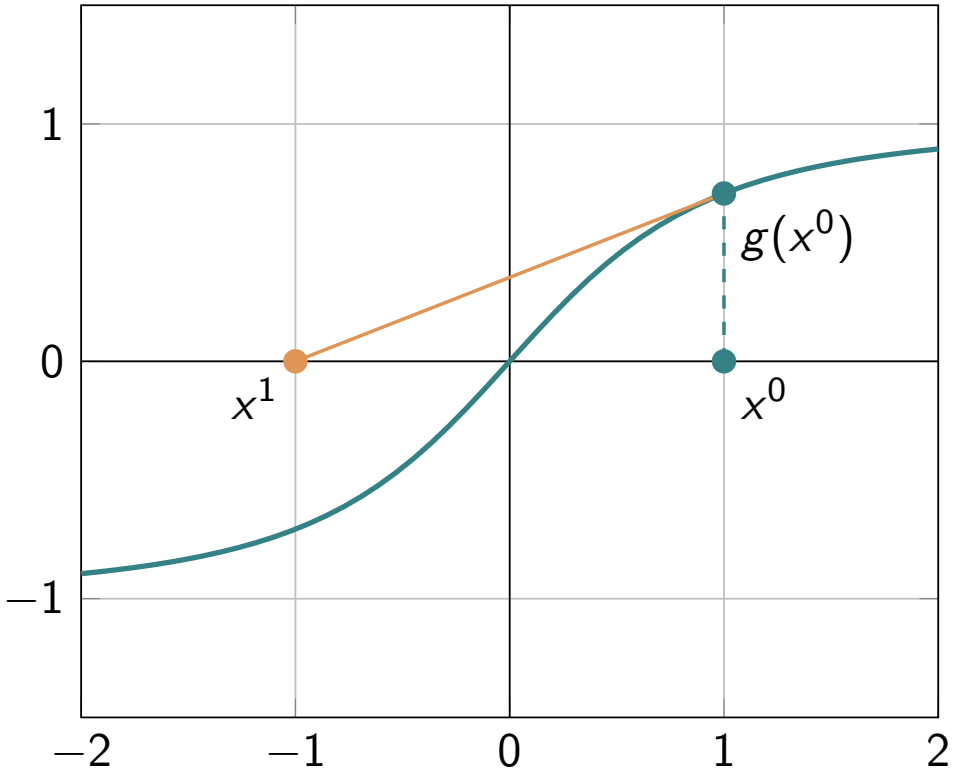
When g is linear, it converges to the solution in 1 iteration.



However . . .

Newton's method may not converge for every initial point.

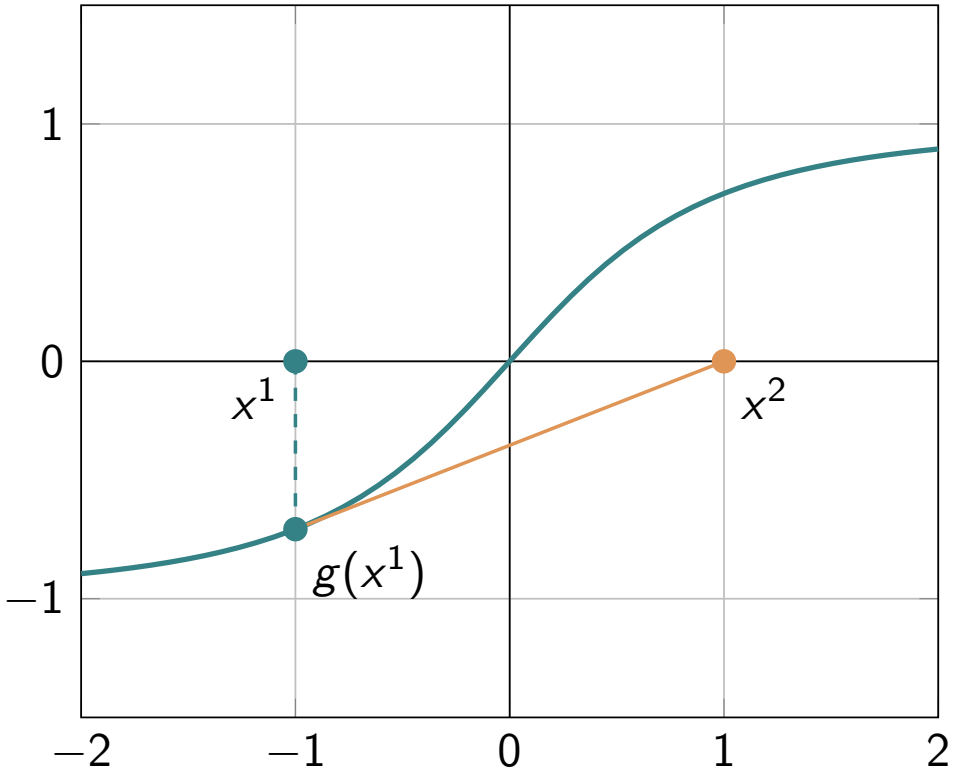
- Consider $g(x) = x/\sqrt{1+x^2}$. It has root $x = 0$.



However . . .

Newton's method may not converge for every initial point.

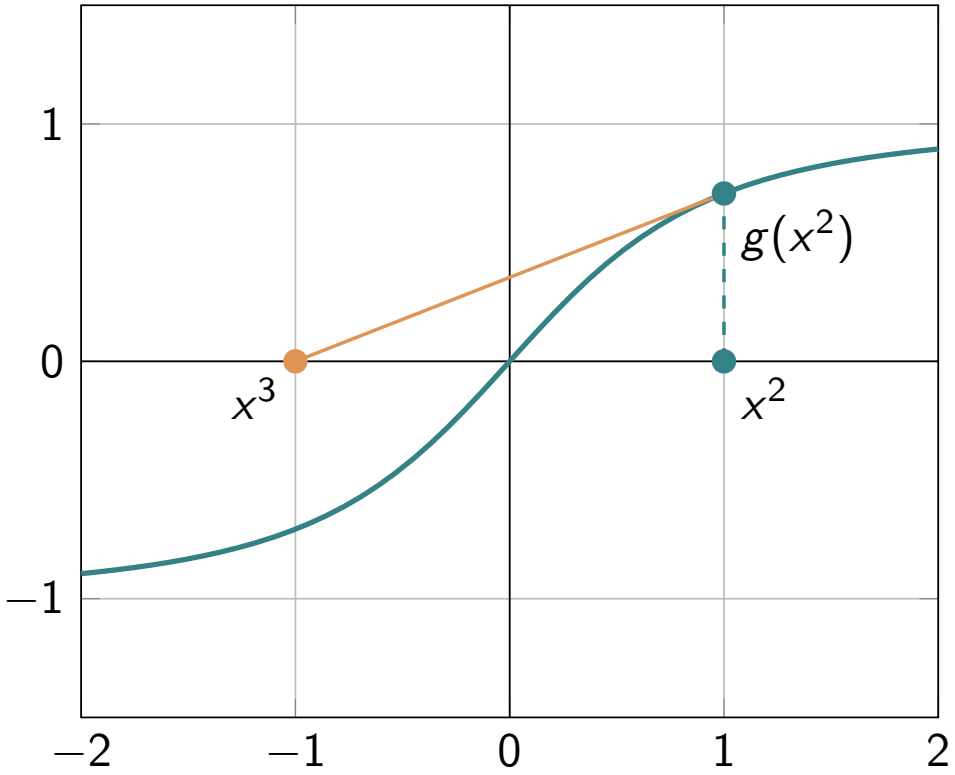
- Consider $g(x) = x/\sqrt{1+x^2}$. It has root $x = 0$.



However . . .

Newton's method may not converge for every initial point.

- Consider $g(x) = x/\sqrt{1+x^2}$. It has root $x = 0$.



Convergence of Newton's Method (1-D Case)

Theorem: Convergence Newton's Method

If g is twice continuously differentiable and x^* is a root of g at which $g'(x^*) \neq 0$, then provided that $|x^0 - x^*|$ is sufficiently small, the sequence generated by the Newton iterations:

$$x^{k+1} = x^k - \frac{g(x^k)}{g'(x^k)}$$

will satisfy

$$|x^{k+1} - x^*| \leq C|x^k - x^*|^2$$

with $C = \frac{1}{2}\sup_{x\in\mathbb{I}} |g''(x)| \cdot \sup_{x\in\mathbb{I}} \left(\frac{1}{|g'(x)|}\right)$, where $\mathbb{I}$ is the interval $\{x : |x - x^*| \leq |x^0 - x^*|\}$.

- We call this convergence speed quadratic convergence.

Linear Convergence vs Quadratic Convergence

The gradient descent method had a linear convergence rate:

$$|x^{k+1} - x^*| \leq \eta \cdot |x^k - x^*|.$$

Newton's method has quadratic convergence rate:

$$|x^{k+1} - x^*| \leq C \cdot |x^k - x^*|^2.$$

Linear Convergence vs Quadratic Convergence

The gradient descent method had a linear convergence rate:

$$|x^{k+1} - x^*| \leq \eta \cdot |x^k - x^*|.$$

Newton's method has quadratic convergence rate:

$$|x^{k+1} - x^*| \leq C \cdot |x^k - x^*|^2.$$

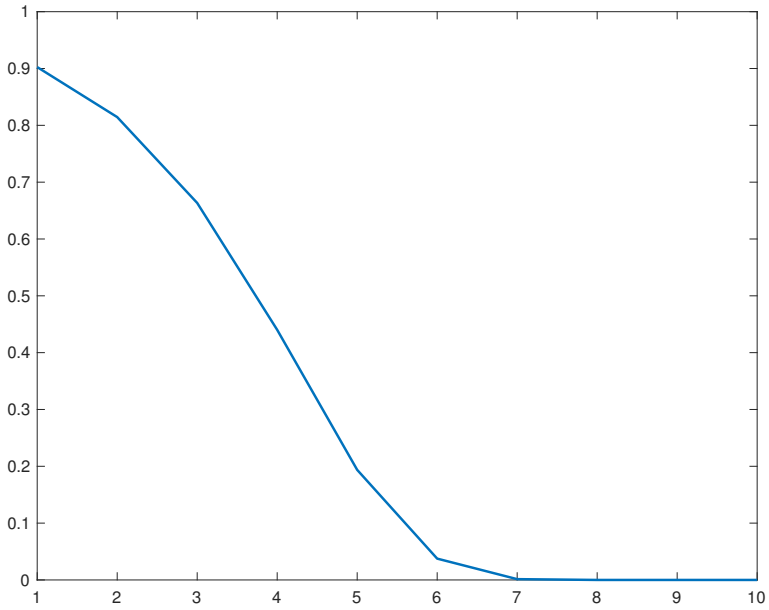
Example:

With $\eta = C = 0.5$ and $|x^0 - x^*| = 0.5$. Then:

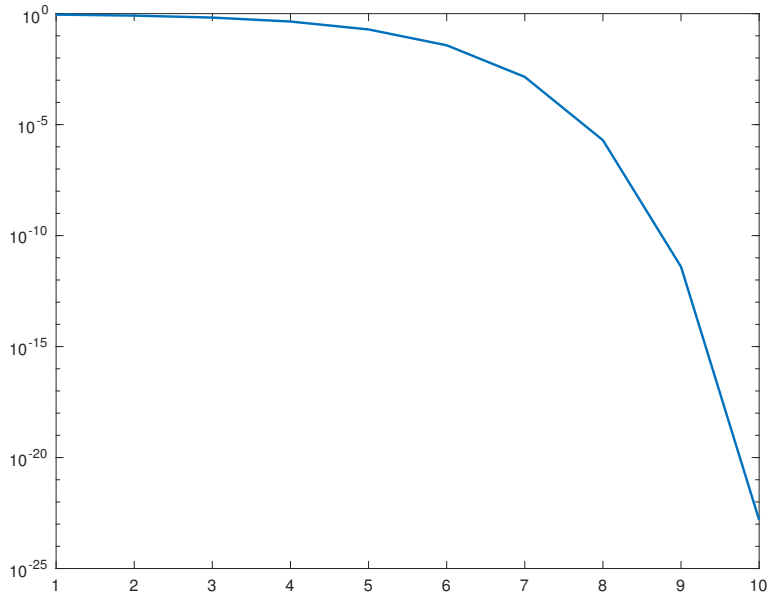
Iteration	1	2	3	5
Gradient (linear conv.)	0.25	0.125	0.063	0.031
Newton (quadratic conv.)	0.125	0.0078	3×10^{-5}	1×10^{-19}

In order to achieve 1×10^{-19}, Newton's method needs 5 iterations, while the gradient method would require 64 iterations.

Illustration of Quadratic Convergence



(a) Plot of $\{0.95^{2^k}\}_k$



(b) Logarithmic plot of $\{0.95^{2^k}\}_k$

- The logarithmic plot in (b) is similar to a quadratic function that opens downward.
- Log of log (quadratic log) will have a line behavior.

Back to the Optimization Problem

We set $g(x) = f'(x)$, where $f(x)$ is the function we want to minimize. In terms of f, the Newton iteration can written as:

$$x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}.$$

Back to the Optimization Problem

We set $g(x) = f'(x)$, where $f(x)$ is the function we want to minimize. In terms of f, the Newton iteration can written as:

$$x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}.$$

- Under proper conditions, the sequence $\{x^k\}_k$ converges to a stationary point of f.
- When f is convex, it converges to the global minimizer (under appropriate assumptions).

Another Interpretation of Newton's Method

The second order Taylor expansion of f at x^k:

$$f(x) \approx f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2} f''(x^k)(x - x^k)^2.$$

Another Interpretation of Newton's Method

The second order Taylor expansion of f at x^k:

$$f(x) \approx f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2} f''(x^k)(x - x^k)^2.$$

The minimizer of of this quadratic function (if $f''(x^k) > 0$) is:

$$x^k - f'(x^k)/f''(x^k)$$

which is exactly the next iterate in Newton's method.

Another Interpretation of Newton's Method

The second order Taylor expansion of f at x^k:

$$f(x) \approx f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2} f''(x^k)(x - x^k)^2.$$

The minimizer of of this quadratic function (if $f''(x^k) > 0$) is:

$$x^k - f'(x^k)/f''(x^k)$$

which is exactly the next iterate in Newton's method.

Interpretation:

- Newton's method builds a quadratic approximation of f locally. The Newton step then is the minimizer of this model.
- If the original objective function is quadratic, then Newton's method converges in one step.

Connection to the Gradient Descent Method

One Newton step is given by:

$$x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}$$

A gradient descent step is given by:

$$x^{k+1} = x^k - \alpha f'(x^k)$$

Connection to the Gradient Descent Method

One Newton step is given by:

$$x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}$$

A gradient descent step is given by:

$$x^{k+1} = x^k - \alpha f'(x^k)$$

Observation:

- In the 1-D case, Newton's method simply specifies a specific step size in the gradient method.
- In the high-dimensional case, however, Newton's method will use different the search directions.

Newton's Method — in $\mathbb{R}^n$

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top (\boldsymbol{x} - \boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k)$$

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top(\boldsymbol{x}-\boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x}-\boldsymbol{x}^k)^\top \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x}-\boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Newton's Method in High Dimensional Case

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

The second-order Taylor expansion:

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}^k) + \nabla f(\boldsymbol{x}^k)^\top(\boldsymbol{x}-\boldsymbol{x}^k) + \frac{1}{2}(\boldsymbol{x}-\boldsymbol{x}^k)^\top\nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x}-\boldsymbol{x}^k)$$

Newton's Method Iteration:

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

This corresponds to the minimizer of the quadratic approximation, provided $\nabla^2 f(\boldsymbol{x}^k)$ is positive definite.

Search Direction (Newton's Direction):

$$\boldsymbol{d}^k = -(\nabla^2 f(\boldsymbol{x}^k))^{-1}\nabla f(\boldsymbol{x}^k).$$

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Using a Taylor expansion at $\boldsymbol{x}^k$, we have

$$\nabla f(\boldsymbol{x}) \approx \nabla f(\boldsymbol{x}^k) + \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k) =: q_k(\boldsymbol{x}).$$

Newton's Method in High Dimensional Case

We can also consider the nonlinear equation $\nabla f(\boldsymbol{x}) = \mathbf{0}$.

Using a Taylor expansion at $\boldsymbol{x}^k$, we have

$$\nabla f(\boldsymbol{x}) \approx \nabla f(\boldsymbol{x}^k) + \nabla^2 f(\boldsymbol{x}^k)(\boldsymbol{x} - \boldsymbol{x}^k) =: q_k(\boldsymbol{x}).$$

The solution to $q_k(\boldsymbol{x}) = \mathbf{0}$ is (when$\nabla^2 f(\boldsymbol{x})$ is invertible)

$$\boldsymbol{x} = \boldsymbol{x}^k - (\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k).$$

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

means $\nabla^2 f(x) - \mu I$ is PSD. It's satisfied if f is μ-strongly convex.

Hessian is Lipschitz smooth.

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

Let $\{\boldsymbol{x}^k\}_k$ be the sequence generated by Newton's method, and let $\boldsymbol{x}^*$ be the unique minizer. Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L}{2\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$$

quadratic convergence

Theorem: Convergence of Newton's Method

Let f be a twice continuously differentiable function over $\mathbb{R}^n$. Assume that

- there exists $\mu > 0$ that $\nabla^2 f(\boldsymbol{x}) \succeq \mu \boldsymbol{I}$ for any $\boldsymbol{x}$,
- there exists $L > 0$ that $\|\nabla^2 f(\boldsymbol{x}) - \nabla^2 f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for all $\boldsymbol{x}, \boldsymbol{y}$.

Let $\{\boldsymbol{x}^k\}_k$ be the sequence generated by Newton's method, and let $\boldsymbol{x}^*$ be the unique minizer. Then

$$\|\boldsymbol{x}^{k+1} - \boldsymbol{x}^*\| \leq \frac{L}{2\mu}\|\boldsymbol{x}^k - \boldsymbol{x}^*\|^2$$

In addition, if $\|\boldsymbol{x}^0 - \boldsymbol{x}^*\| \leq \frac{\mu}{L}$, then

$$\|\boldsymbol{x}^k - \boldsymbol{x}^*\| \leq \frac{2\mu}{L}\left(\frac{1}{2}\right)^{2^k}$$

.

Convergence Newton's Method:

- "Hessian uniformly positive definite + Lipschitz continuous" and "$\boldsymbol{x}^0$ close to $\boldsymbol{x}^*$"

$$\Longrightarrow \quad \{\boldsymbol{x}^k\}_k \text{ converges quadratically to } \boldsymbol{x}^*$$

- **Proof:** See Theorem 5.2 in Beck: "Introduction to Nonlinear Optimization".

Newton's Step as a Descent Step

Recap: A vector $\boldsymbol{d}$ is a descent direction if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

In Newton's method, we have

$$\boldsymbol{d} = -(\nabla^2 f(\boldsymbol{x}))^{-1} \nabla f(\boldsymbol{x}).$$

Newton's Step as a Descent Step

Recap: A vector $\boldsymbol{d}$ is a descent direction if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

In Newton's method, we have

$$\boldsymbol{d} = -(\nabla^2 f(\boldsymbol{x}))^{-1} \nabla f(\boldsymbol{x}).$$

Then, it holds that:

$$\nabla f(\boldsymbol{x})^\top \boldsymbol{d} = -\nabla f(\boldsymbol{x})^\top (\nabla^2 f(\boldsymbol{x}))^{-1} \nabla f(\boldsymbol{x}).$$

Newton's Step as a Descent Step

Recap: A vector $\boldsymbol{d}$ is a descent direction if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

In Newton's method, we have

$$\boldsymbol{d} = -(\nabla^2 f(\boldsymbol{x}))^{-1} \nabla f(\boldsymbol{x}).$$

Then, it holds that:

$$\nabla f(\boldsymbol{x})^\top \boldsymbol{d} = -\nabla f(\boldsymbol{x})^\top (\nabla^2 f(\boldsymbol{x}))^{-1} \nabla f(\boldsymbol{x}).$$

- If f is strongly convex, then $\nabla^2 f(\boldsymbol{x})$ is positive definite, and therefore $\nabla^2 f(\boldsymbol{x})$ is invertible and $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$. when $\nabla f(x)$ is not 0

Step Length

As we said earlier, Newton's method may not converge unless the initial point is sufficiently close to a solution.

Step Length

As we said earlier, Newton's method may not converge unless the initial point is sufficiently close to a solution.

One way to ensure convergence is to use a step size parameter α_k

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$$

where $\boldsymbol{d}^k = -(\nabla^2 f(\boldsymbol{x}^k))^{-1} \nabla f(\boldsymbol{x}^k)$ is Newton's direction.

- We can use backtracking line search to determine α_k.

Complete Procedure of Newton's Method

(Damped) Newton Method

1. Initialization: Select an initial point $\boldsymbol{x}^0 \in \mathbb{R}^n$.

For $k = 0, 1, \ldots$:

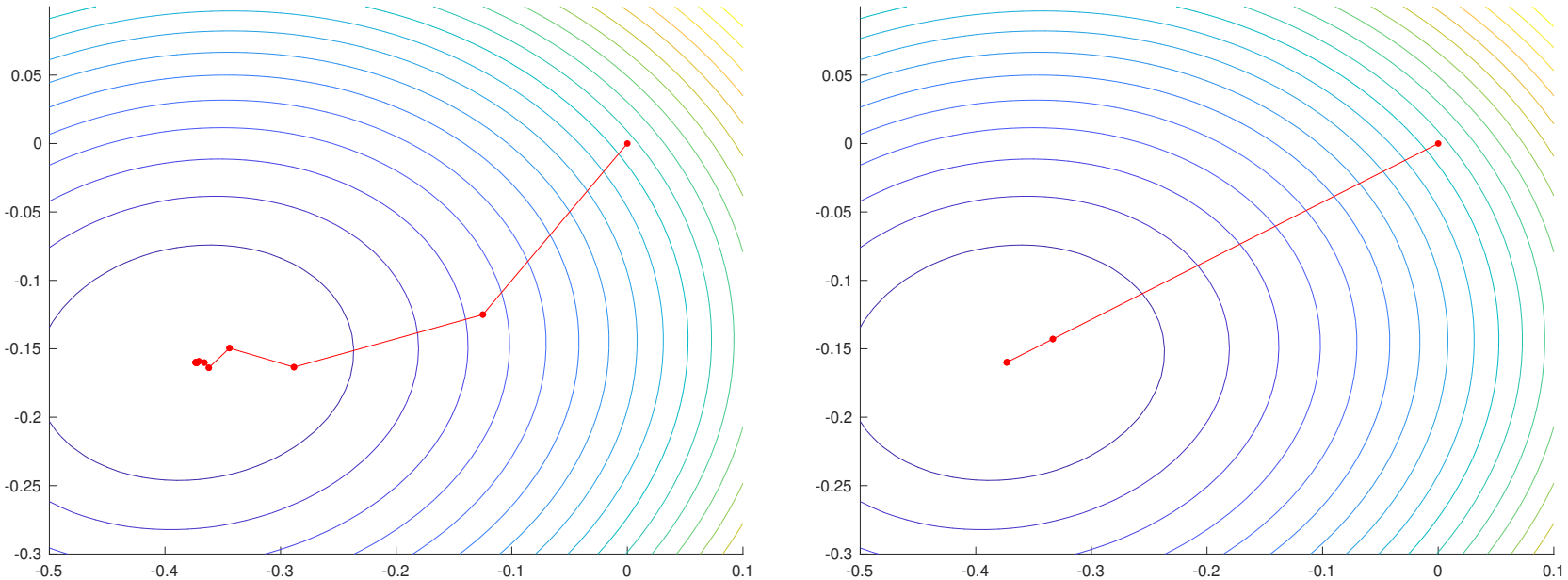
2. If $\|\nabla f(\boldsymbol{x}^k)\| \leq \varepsilon$, then STOP and $\boldsymbol{x}^k$ is the output.
3. Compute the Newton's direction $\boldsymbol{d}^k$ which is the solution of the linear system
$$\nabla^2 f(\boldsymbol{x}^k)\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k).$$
4. Newton's method: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \boldsymbol{d}^k$
Damped Newton's method: $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha_k \boldsymbol{d}^k$, where $\alpha_k \leq 1$ is decided by the step size strategy, such as backtracking line search.

Example I

Minimize

$$f(\boldsymbol{x}) = \exp(x_1 + x_2) + x_1^2 + 3x_2^2 - x_1 x_2$$

using the gradient method and Newton's method.



- **Left:** Gradient method (Armijo). **Right:** Newton's method.

Example — II

We again consider the optimization problem

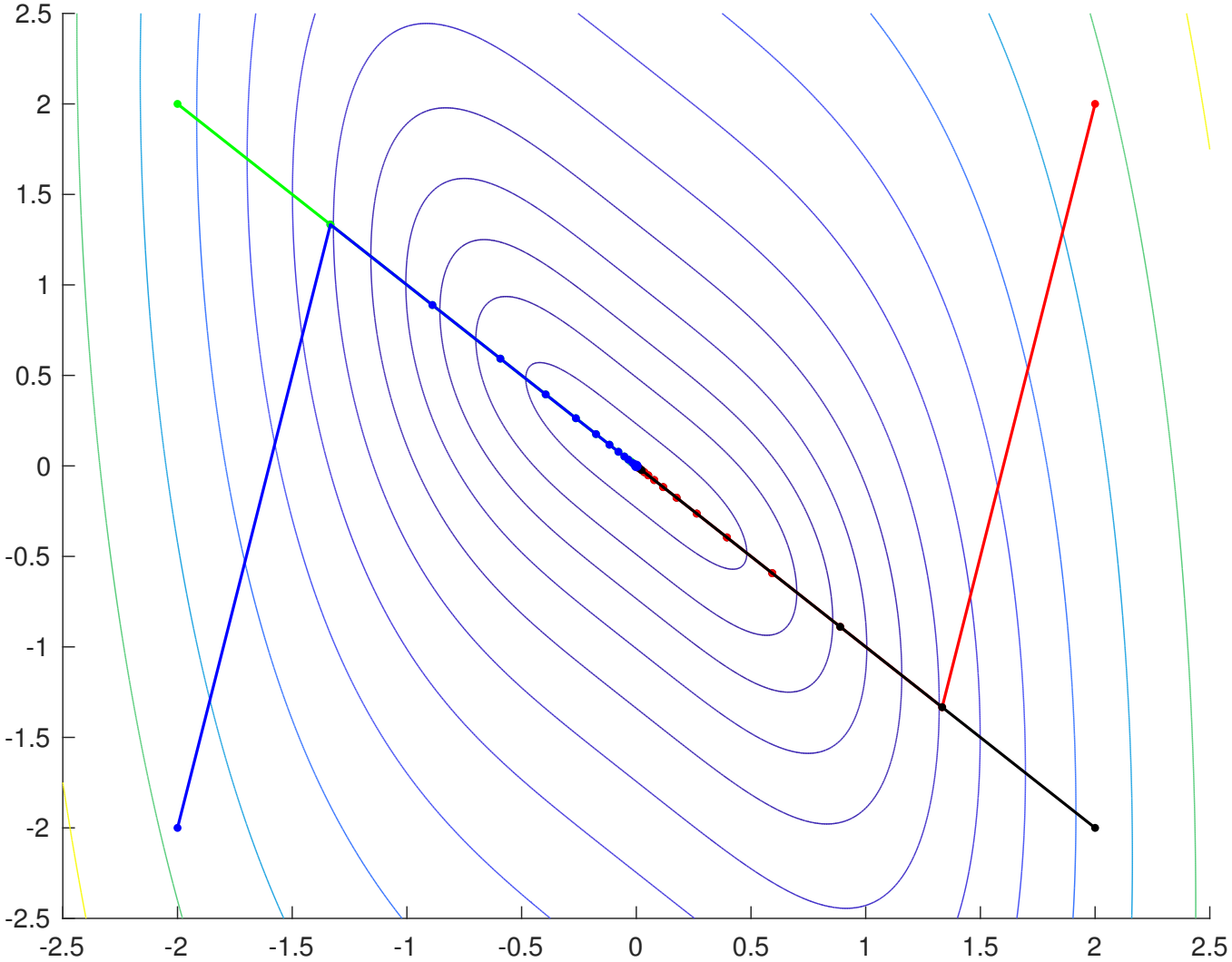
$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \; f(\boldsymbol{x}) = x_1^4 + x_1^2 + x_2^2 + 2x_1x_2 + \frac{1}{4}.$$

- This problem has only one stationary point: $\boldsymbol{x}^* = (0,0)^\top$ (which is a strict global minimizer).

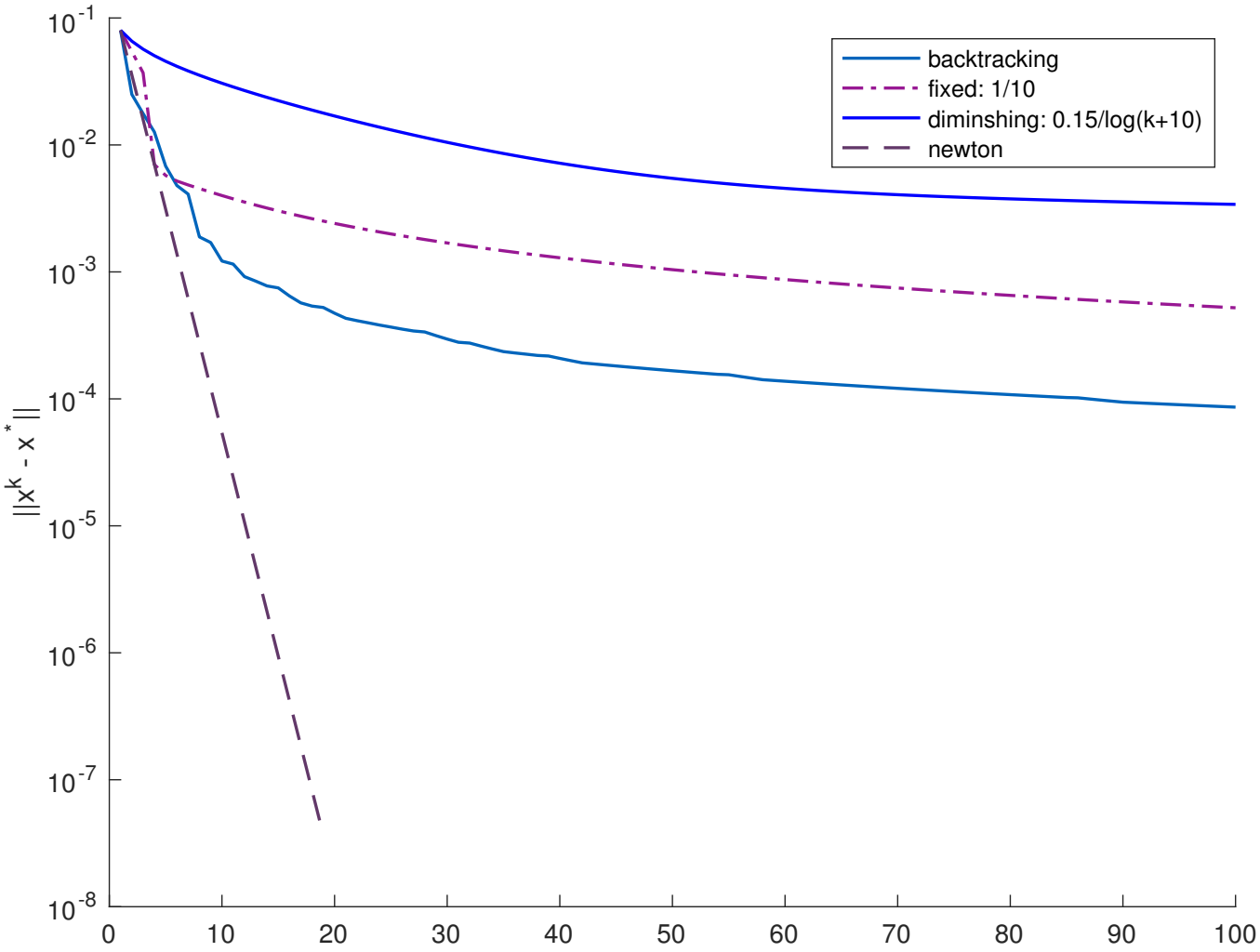
We use the Newton method with the following parameters:

- $\gamma_1 = 10^{-6}$, $\gamma_2 = 0.1$ and $\gamma = 0.1$, $\sigma = 0.5$ (for backtracking).

Solution Paths: Newton's Method



Convergence: Comparison



Newton's Method is Fast, But ...

Challenges of Newton's Method:

- **Hessian Computation:** Requires $\nabla^2 f(\boldsymbol{x})$ at each iteration, computationally expensive for high-dimensional problems.
- **Memory:** Storing the Hessian matrix requires $O(n^2)$ space, significantly more than the $O(n)$ for gradient descent.
- **Matrix Operations:** Each iteration involves costly matrix inversions or solving linear systems.

Newton's Method is Fast, But ...

Challenges of Newton's Method:

- **Hessian Computation:** Requires $\nabla^2 f(\boldsymbol{x})$ at each iteration, computationally expensive for high-dimensional problems.
- **Memory:** Storing the Hessian matrix requires $O(n^2)$ space, significantly more than the $O(n)$ for gradient descent.
- **Matrix Operations:** Each iteration involves costly matrix inversions or solving linear systems.

Partial Solutions

- **Quasi-Newton Method:** Instead of calculating $\nabla^2 f(\boldsymbol{x})$ explicitly, approximate it using past gradients.
- **Inexact Newton Methods:** Solve the Newton system approximately.
- Compute Hessian ~~inverse~~ only periodically.

use $d_k = -[\nabla^2 f(x_l)]^{-1} \nabla f(x_k)$ where l is an old iteration.

Trade-off between Newton and Gradient Method

- Gradient method is simple and reasonably fast in each iteration and needs less storage and computation.
- Newton's method takes much less iterations to converge. However, each iteration is more expensive in both computation and storage space.

There are many numerical improvements that one can make but the above main tradeoff always remains:

- Whether the gradient method or Newton's method should be used depends largely on the problem at hand.

Today's lecture

- Newton's method

Next lecture

- Algorithms for Constrained Problems

Questions?